Homework 5

Differential Geometry

Due on May 24

Only **handwritten solutions** will be accepted; typed solutions are **not** allowed. In all problems, you must show your reasoning and derivations clearly, step by step. Although it is easy to find answers using online tools, the main purpose of this homework is to deepen your understanding through careful, hands-on problem solving. Working through the details yourself is an essential part of learning the material.

Problem 1. Let us define

$$S^3 = \left\{(x^0, x^1, x^2, x^3) \in \mathbb{R}^4 \;\middle|\; \textstyle\sum_{i=0}^{3}(x^i)^2 = 1\right\}$$

and

$$S^1 = \left\{(x^0, x^1, x^2, x^3) \in \mathbb{R}^4 \;\middle|\; (x^0)^2 + (x^1)^2 = 1\right\}.$$

Show that $S^3 \setminus S^1$ is homotopic to S^1.

Problem 2 (Fundamental theorem of algebra). Define $f\colon \mathbb{C} \to \mathbb{C}$ by

$$f(z) = z^n + a_1 z^{n-1} + \cdots + a_n, \quad n \geq 1.$$

Writing $z = x + iy$, define the one-form

$$\omega = \operatorname{Im}\frac{dz}{z} = \frac{-y\,dx}{x^2+y^2} + \frac{x\,dy}{x^2+y^2}.$$

(a) Show that

$$\frac{1}{2\pi}\int_{C_R} f^*\omega = n,$$

where C_R is the circle of sufficiently large radius R. (*Hint*: construct a homotopy between f and $g(z) = z^n$.)

(b) If f has no zero inside C_R, show via Stokeś theorem that

$$\frac{1}{2\pi}\int_{C_R} f^*\omega = 0.$$

Problem 3. Find all Poincaré dual pairs (non-trivial intersection pairings) in the real-valued homology group $H_\ell(\Sigma_g; \mathbb{R})$ of the Riemann surface Σ_g of genus g.

Problem 4. Find the fundamental group $\pi_1(\Sigma_g)$ of the Riemann surface of genus g.